

Markscheme

Mathematics: analysis and approaches
Practice question bank

152. (a)

Consider $z = \sqrt{3} - i$.

Express z in exponential form, and hence find z^4 in the form $a + bi$, correct to 3 significant figures.

$$r = \sqrt{3 + 1} = 2, \theta = -\frac{\pi}{6}, \text{ so } z = 2e^{-i\pi/6} \quad (M1) \text{ A1}$$

$$z^4 = 2^4 e^{-i2\pi/3} = 16e^{-i2\pi/3} \quad (M1)$$

$$= 16(\cos(-120^\circ) + i \sin(-120^\circ)) \quad (M1)$$

$$\approx -8 - 13.9i \quad \text{A1}$$

[6 marks]